

CFD_HW_4

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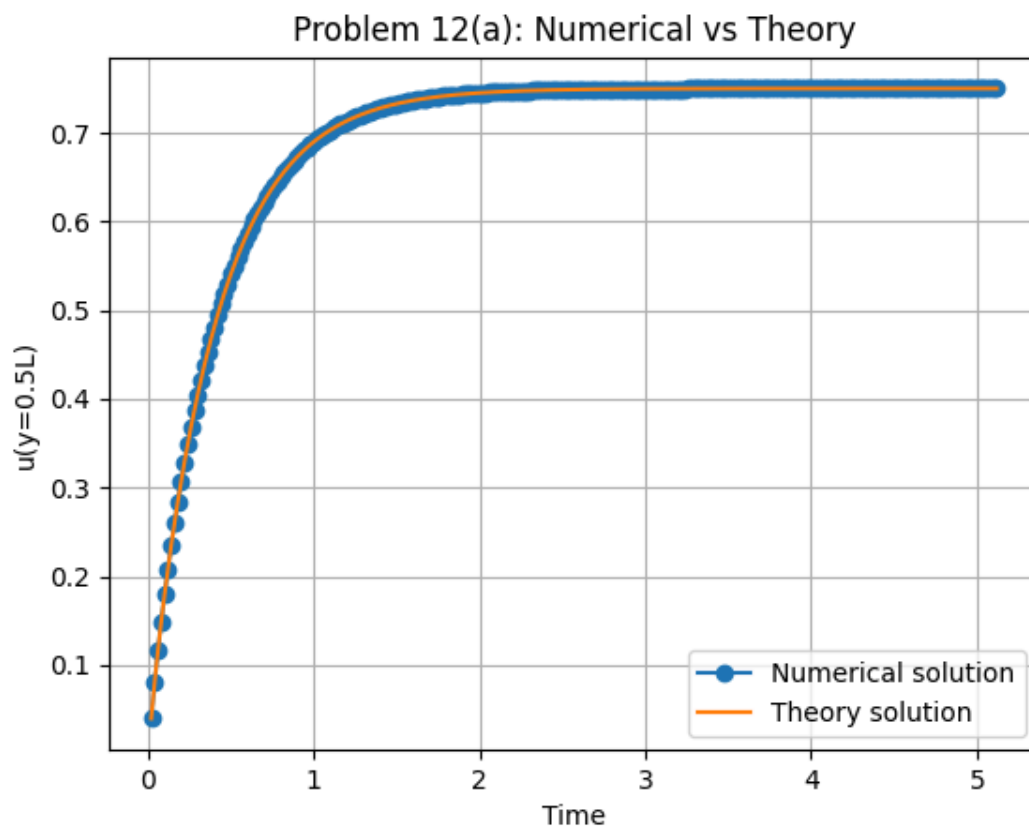
学号：12532299

课程：计算流体力学

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Question 12: Exact Solution of Finite-Difference Scheme

(a)

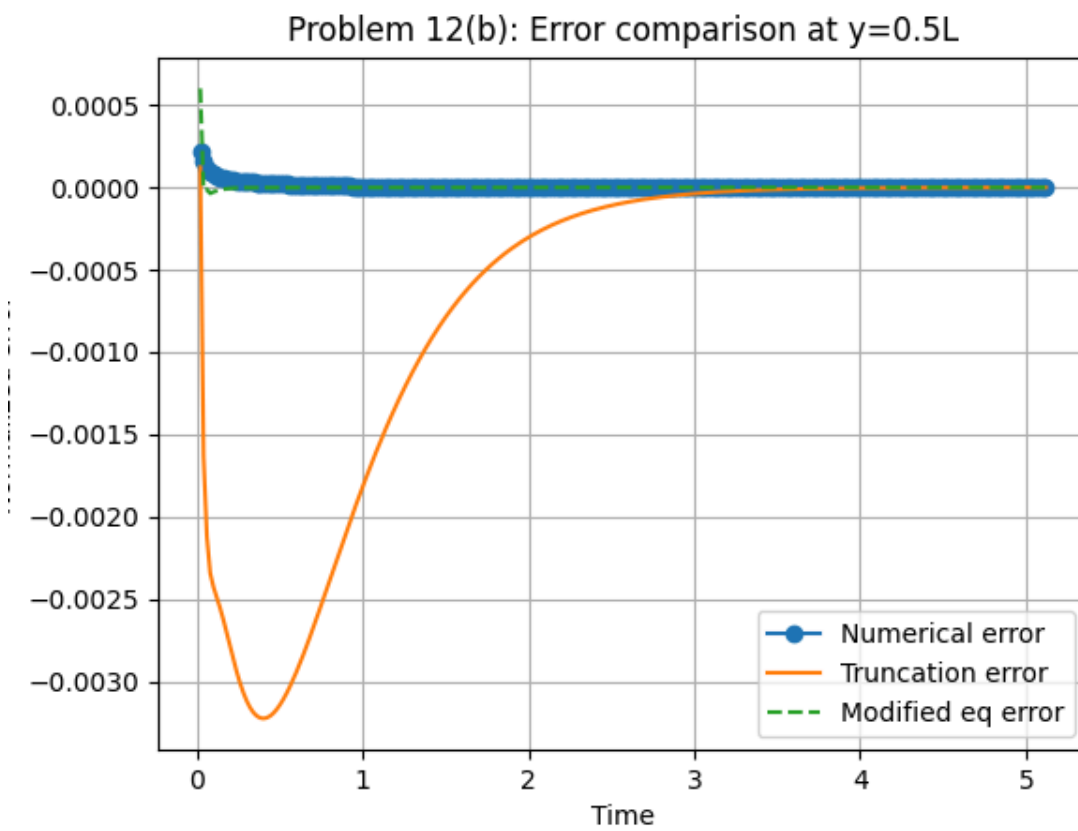


The plots are almost the same between the numerical solution u_j^n and the exact solution of the finite difference scheme.

(b)

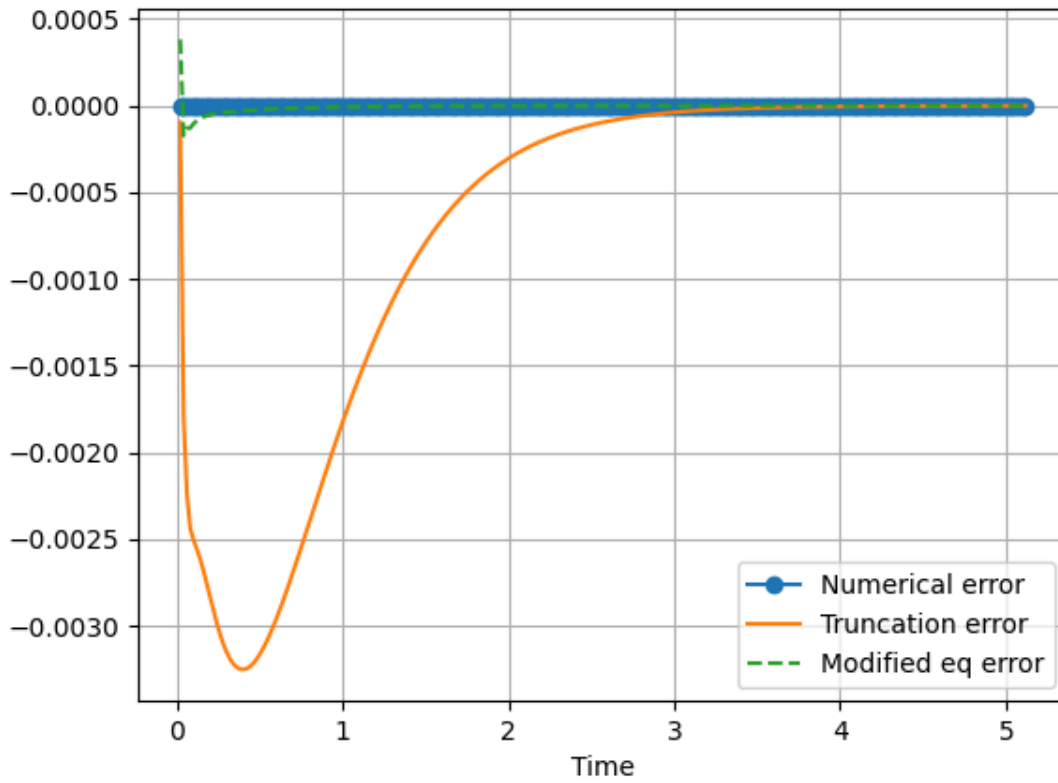
The results are :

$$k_{max} = 5$$



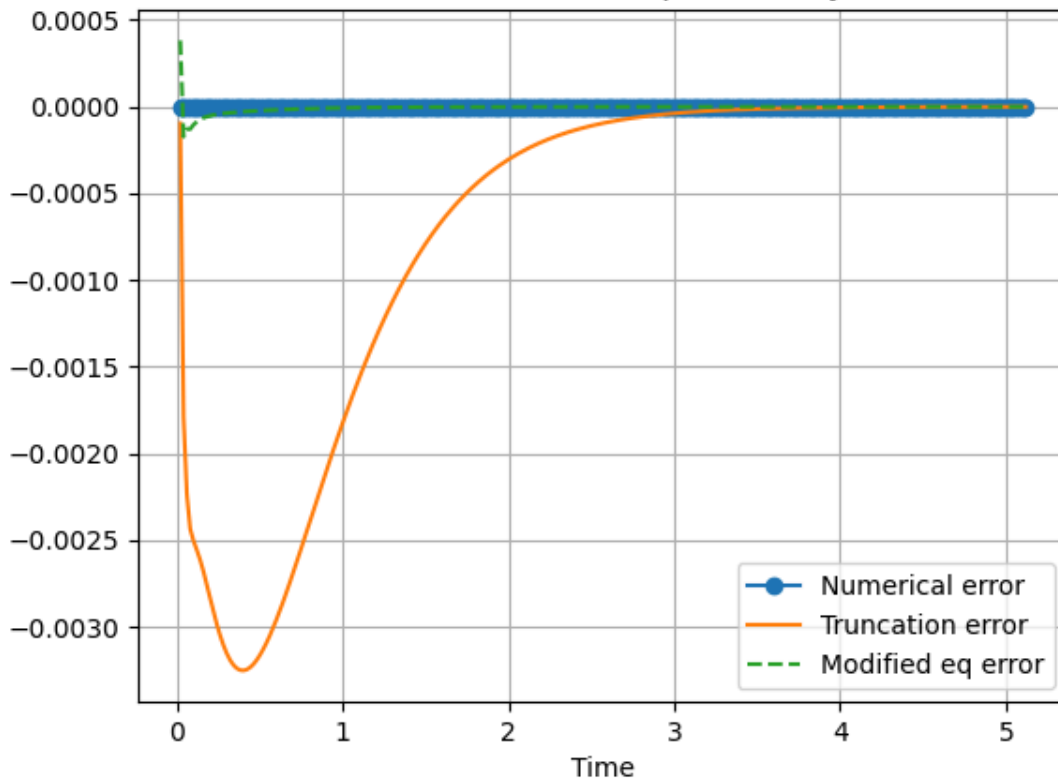
$$k_{max} = 50$$

Problem 12(b): Error comparison at $y=0.5L$



$$k_{max} = 100$$

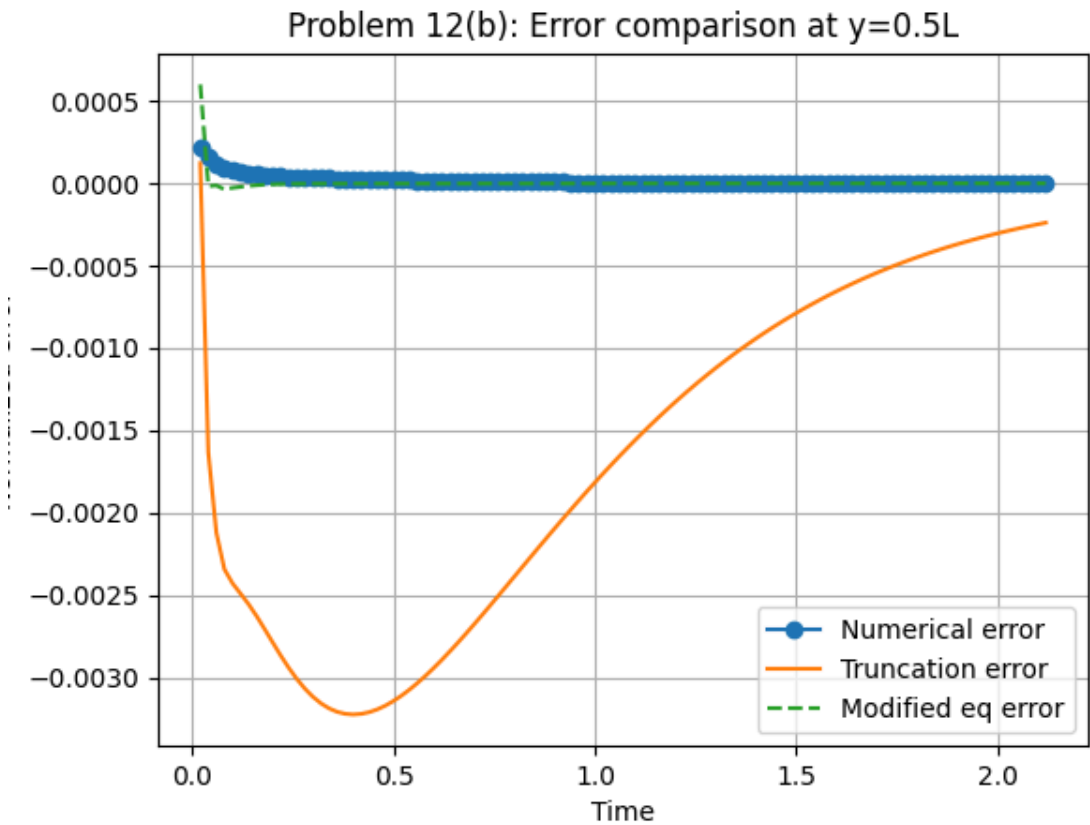
Problem 12(b): Error comparison at $y=0.5L$



The dominant source of error is not from numerical implementation but from the inherent truncation error of the finite-difference scheme.

We can find that at the beginning, different k_{max} choices lead to different results. However, as the time goes, they are go to steady stats as the times go longer.

If we only compute 2 times of specific time:

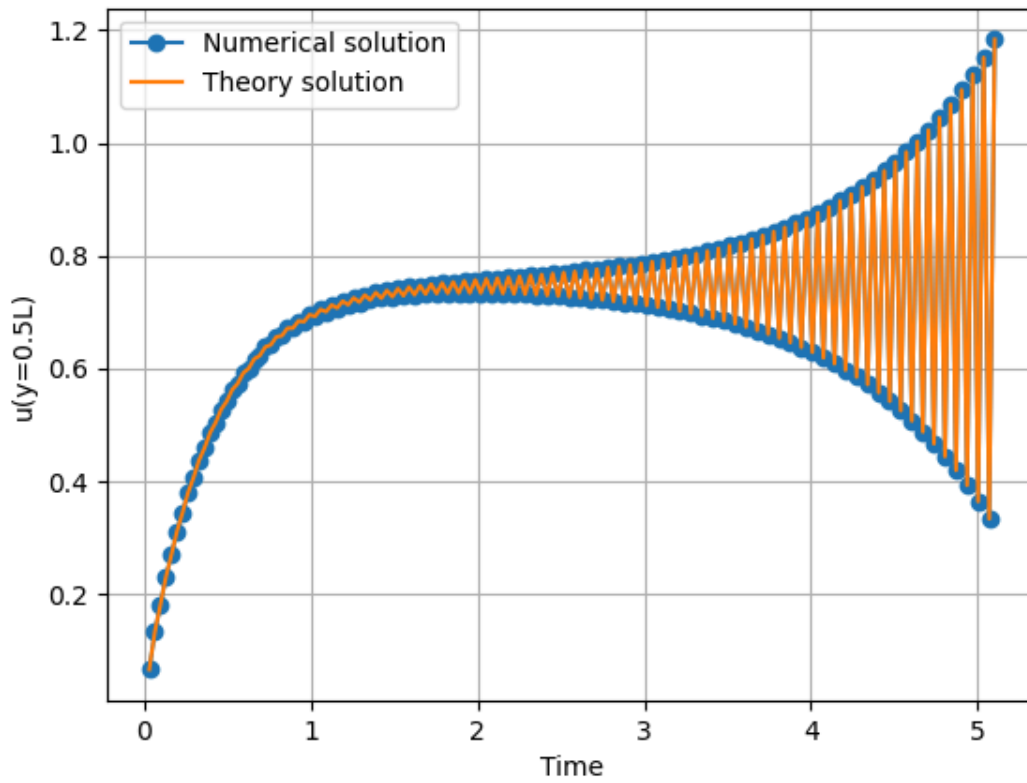


We can find the differences are at about 0 ~ 0.2.

(c)

The results for CFL = 0.53 in (a) are :

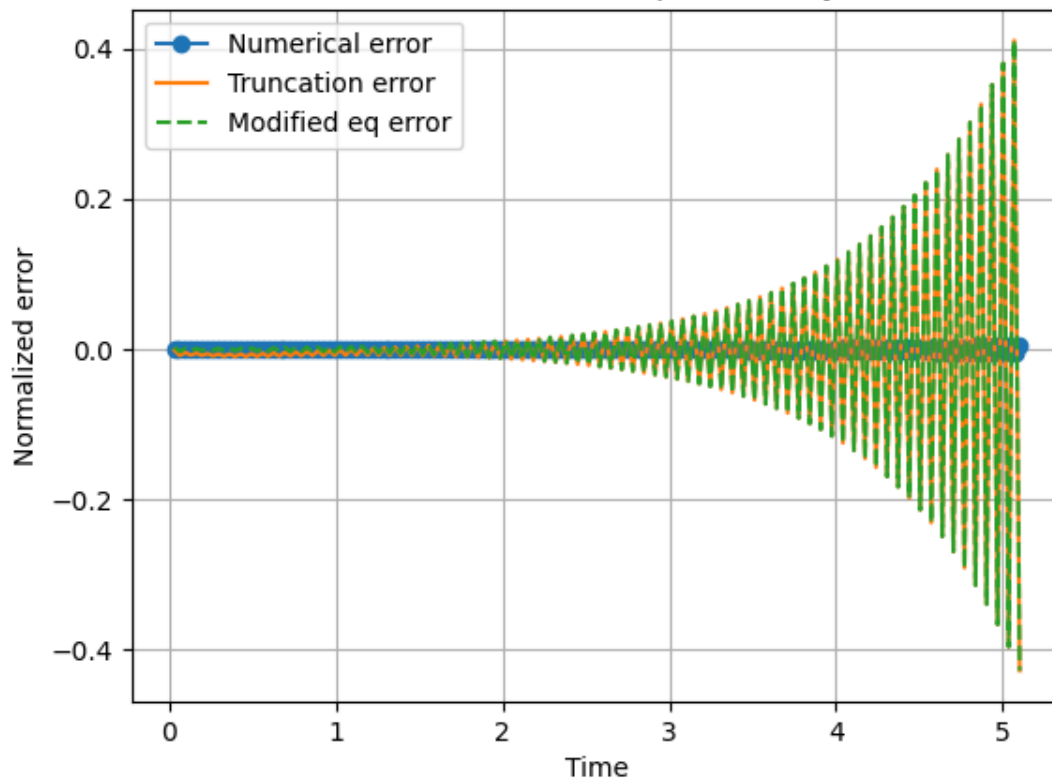
Problem 12(a): Numerical vs Theory



The results for CFL = 0.53 in (b) are :

$$k_{max} = 5$$

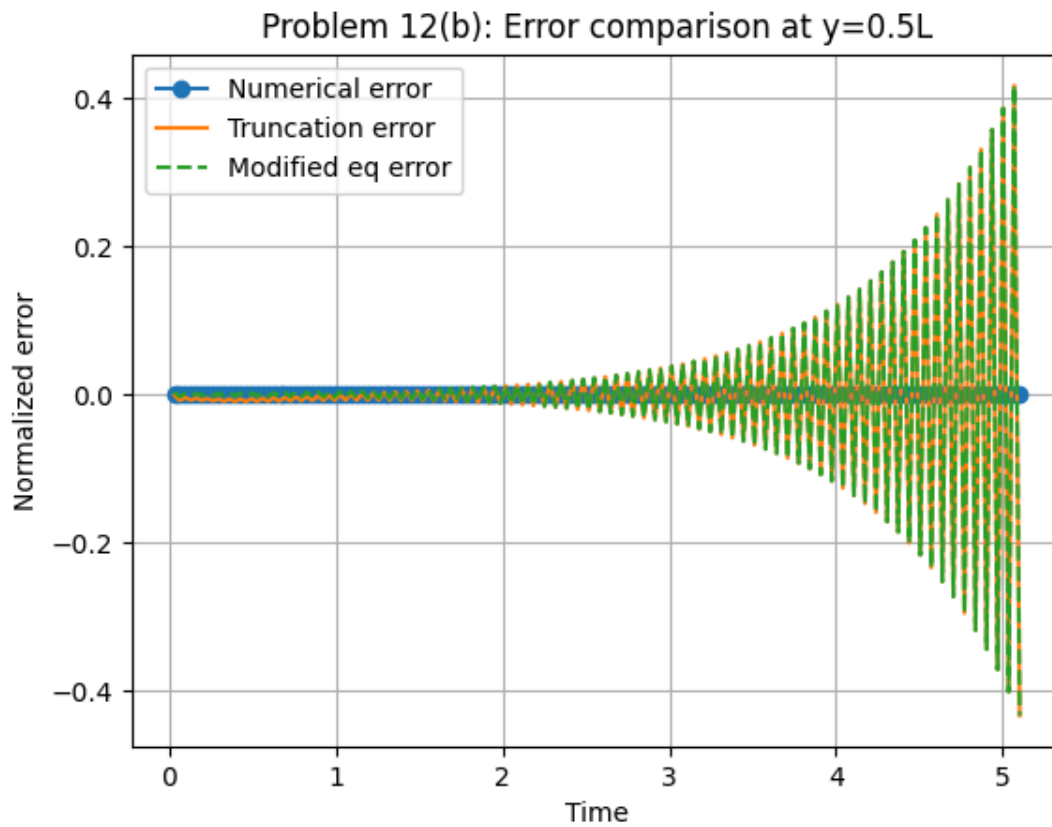
Problem 12(b): Error comparison at $y=0.5L$



$k_{max} = 50$



$k_{max} = 100$



Comments:

1. The numerical solution u_j^n and the exact solution of the finite difference scheme fixed very well, even if $CFL > 0.5$ (unstable). The error between these 2 solutions mostly from the number of k_{max} .
2. Of course, as $CFL > 0.5$, the results become not stable. But the unstable behavior **shows** clearly at about $T > 2.0t$.

Question 13: 1D advection equation

(a)

We consider the 1D advection equation:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

where $a = 1$.

To solve this equation, we apply the method of characteristics. The total derivative of u along a curve $x(t)$ is:

$$\frac{du}{dt} = \frac{\partial u}{\partial t} + \frac{dx}{dt} \frac{\partial u}{\partial x}$$

Comparing this with the governing equation:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

we choose the characteristic curve such that:

$$\frac{dx}{dt} = a = 1$$

Then,

$$\frac{du}{dt} = 0$$

which implies that u remains constant along the characteristic lines.

Solving:

$$\frac{dx}{dt} = 1$$

we obtain:

$$x = t + C \quad \Rightarrow \quad x - t = C$$

Thus, the solution is constant along lines:

$$x - t = \text{constant}$$

Since u is constant along characteristics:

$$u(x, t) = f(x - t)$$

The initial condition is:

$$u(x, 0) = \sin(\pi x) + 0.6 \sin(3\pi x)$$

At $t = 0$:

$$u(x, 0) = f(x)$$

Therefore:

$$f(x) = \sin(\pi x) + 0.6 \sin(3\pi x)$$

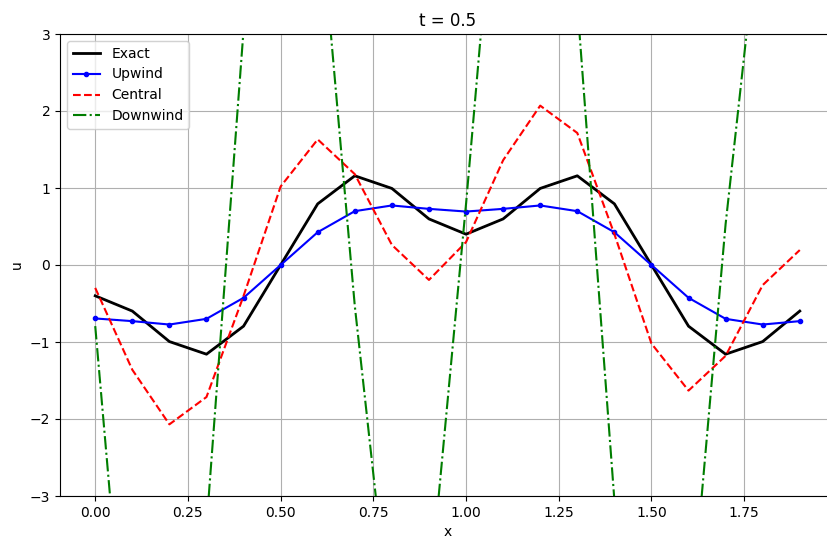
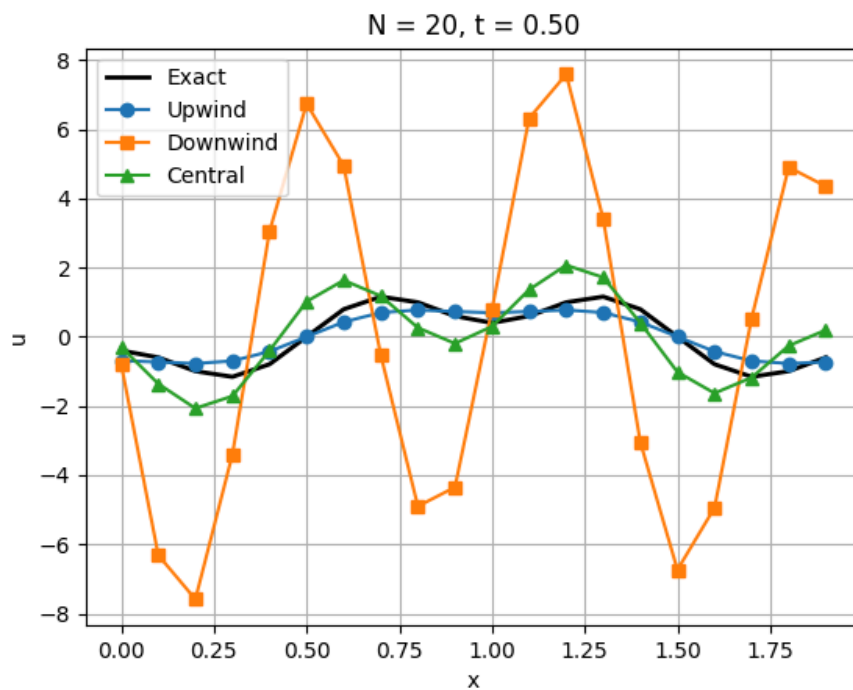
Substituting back:

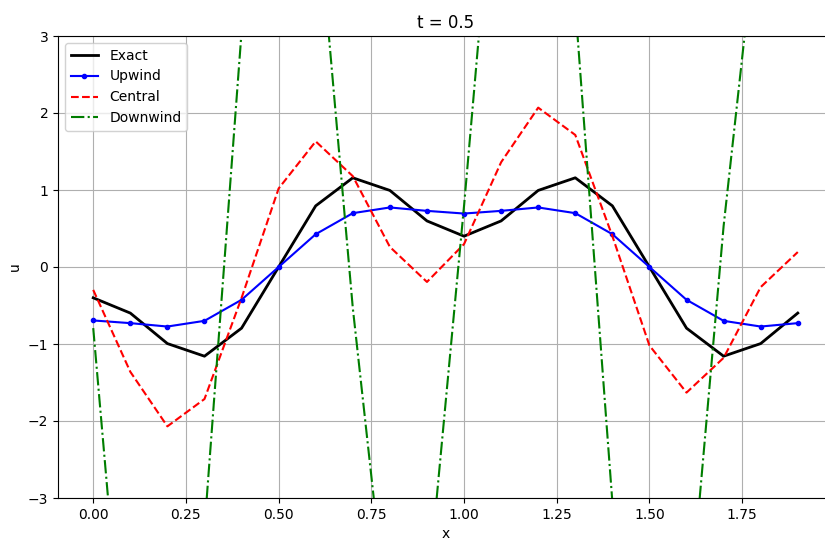
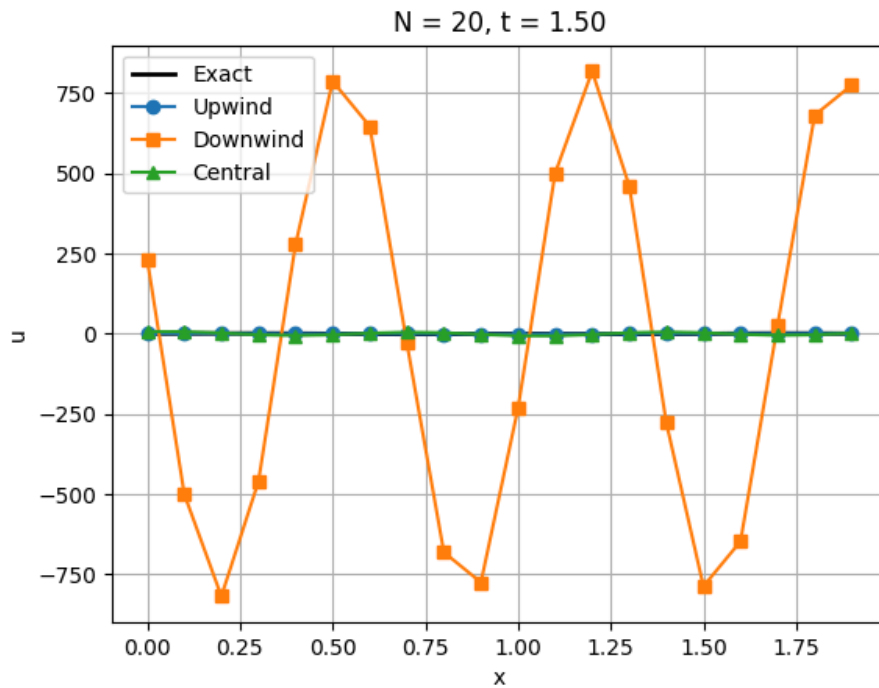
$$u(x, t) = \sin(\pi(x - t)) + 0.6 \sin(3\pi(x - t))$$

The solution represents a wave that propagates to the right with speed $a = 1$ without changing its shape. This is a pure advection process.

(b)

The results in $N = 20$:





(The plots at right limit the y axis to $[-3,3]$)

We can find :

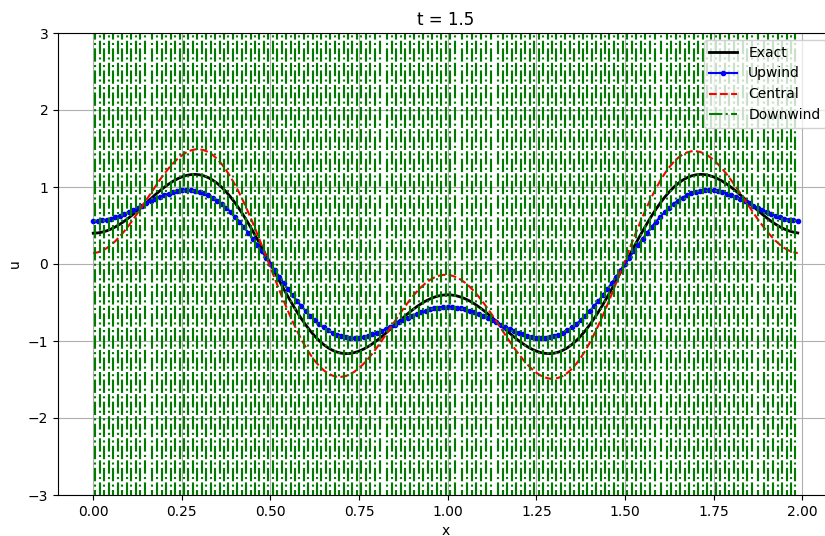
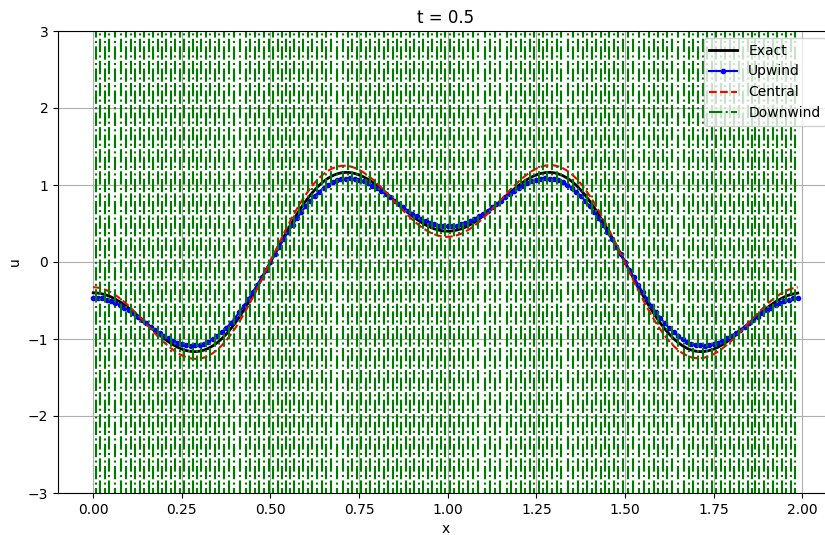
The **downwind scheme diverge** at $t = 0.5s$ (or earlier), as $t = 1.5$ it diverges crazily.

The **central scheme** a little bit **diverges** and has a **phase angle** with the exact answer.

The **upwind scheme** fixed best with the exact answer.

(c)

The results in $N = 160$:



(The plots at right limit the y axis to [-3,3])

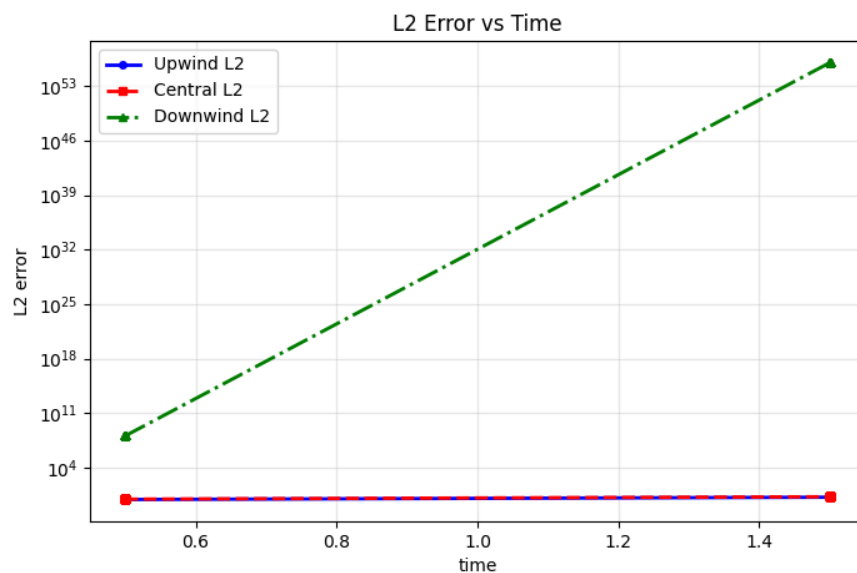
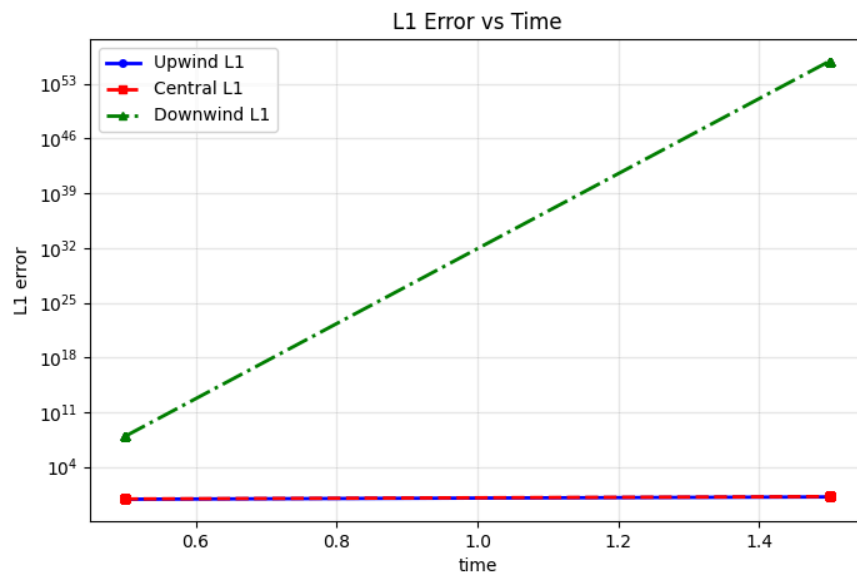
We can find :

The **downwind scheme diverge** at $t = 0.5s$ (or earlier), as $t = 1.5$ it diverges crazily.(same)

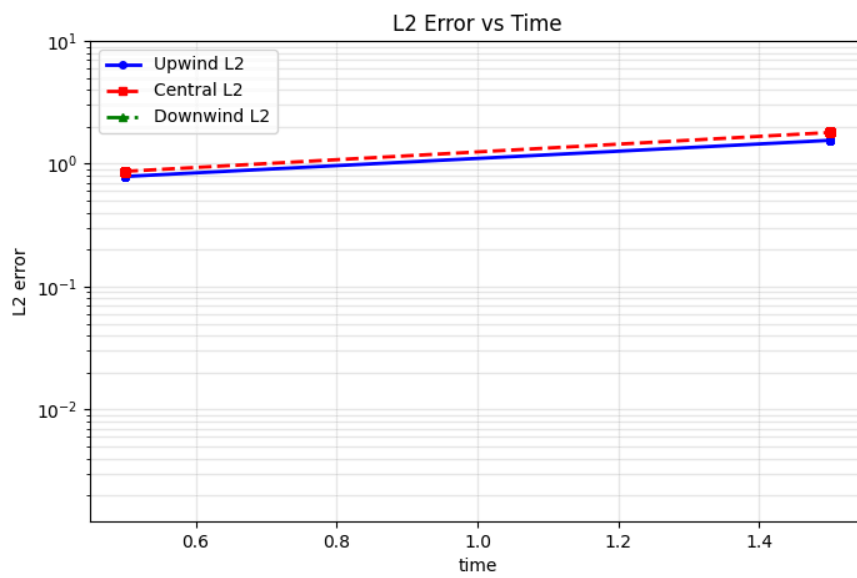
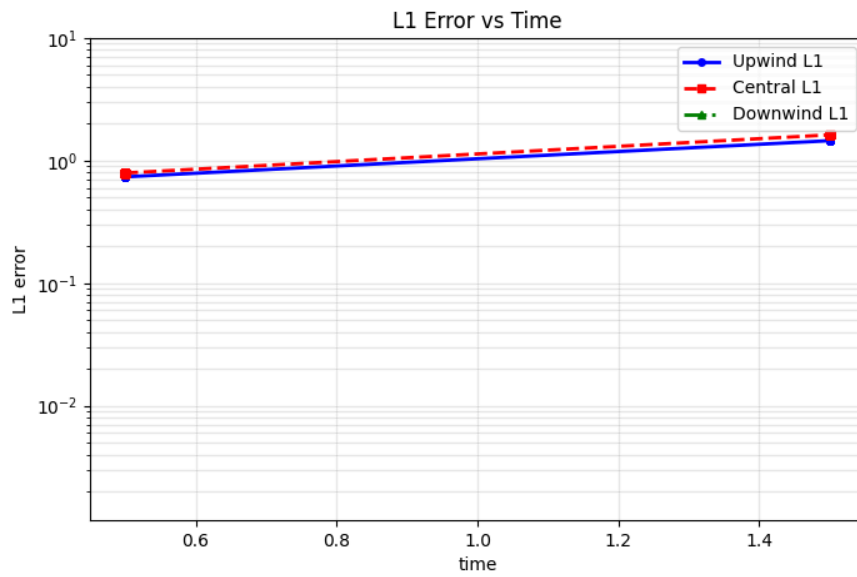
The **central scheme** a little bit **diverges** and has a **phase angle** with the exact answer. (not that clear **phase angle** at $t = 1.5$). And its **amplitude** become **larger**.

The **upwind scheme** fixed best with the exact answer. And its **amplitude** become **smaller**.

(d)



As the plots above : The **downwind scheme** get a bad error progression. While the **central scheme** and the **upwind scheme** seemed not bad, here are the detail for these two methods :



upwind scheme is a little bit better.